

1.1 Fundamentals of Human Computer Interaction (HCI)

ہیومن کمپیوٹر انٹرایکشن (HCI) کا مطلب ہے ٹیکنالوجی کو اس طرح ڈیزائن کرنا کہ وہ انسانوں کے لیے فائدہ مند، آسان اور مؤثر ہو۔

CS XII

Chapter 1: Computer Systems (HCI)

Lecture 01 of 15

1.1 Fundamentals of Human Computer Interaction (HCI)

LEARN IT

Human-Computer Interaction (HCI) is the study and design of technology so that it **serves people**. An HCI system is easy to use, efficient, and responsive to human needs. HCI is also known as **CHI** (Computer-Human Interface) or **MMI** (Man-Machine Interaction). The three names mean the same idea: the meeting point of a person and a machine.

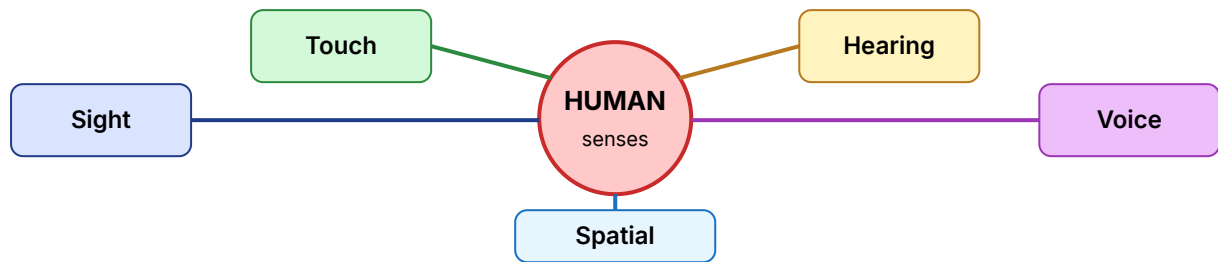
HCI is not only "making a pretty screen". It asks: Can a student in Hyderabad find results on the school portal in 30 seconds? Can a grandmother send JazzCash without help? Can a person standing at an outdoor ATM in bright sunlight read the buttons?

Sensory channels of interaction

Humans and computers communicate through **sensory channels**. A sensory channel is a path that carries information to (or from) one of the human senses. Computers must send and receive signals on the same channels, or the conversation breaks.

Channel	How humans use it	How computers respond
Sight (بصارت)	Look at screens, graphics or videos.	Displays visuals on monitors and screens.
Touch (چھونا)	Tap, type or swipe.	Responds with clicks or vibrations (haptics).
Hearing (سننا)	Listen to sounds or voices.	Provides audio feedback via speakers.
Voice (آواز)	Speak commands (for example, "Alexa").	Processes speech and replies.
Spatial (مکانی)	Move in physical space.	Tracks position or body movement (for example, VR).

Haptics means physical feedback you can feel, such as a phone vibrating when a JazzCash payment succeeds. **Spatial** interaction means the system notices where your body is, as in a VR headset or a Kinect camera.



Five sensory channels meet at the human. HCI design must support the channels the user actually needs.

WORKED EXAMPLE

Worked example: map a JazzCash payment to channels. Sara in Sukkur sends Rs 500 to her brother.

1. **Sight:** she looks at the JazzCash home screen and the amount field.
2. **Touch:** she taps "Send Money" and types the mobile number.
3. **Voice (optional):** she could say "JazzCash, send five hundred rupees" if voice commands are on.
4. **Hearing + haptics:** a beep and a short vibration confirm "Payment successful".

If the app showed no tick, no sound and no vibration, Sara would not know whether the money left. That is an HCI failure, not a banking failure.

REMEMBER IT

Remember the three names: HCI = CHI = MMI. Think "Human meets Machine." Five channels: **S-T-H-V-S** — Sight, Touch, Hearing, Voice, Spatial. "See The Handy Voice in Space."

COMMON MISTAKE

Mistake: Writing "HCI is only about computer hardware" or mixing up Voice with Hearing.

Fix: HCI is about the *design of the conversation* between people and machines. Hearing is what you *listen to* (speakers). Voice is what you *speak* (microphone + speech software).

ہیومن کمپیوٹر انٹرایکشن (HCI) کا مطلب ہے ٹیکنالوجی کو اس طرح ڈیزائن کرنا کہ وہ انسانوں کے لیے فائدہ مند، آسان اور مؤثر ہو۔ اسے CHI یا MMI بھی کہتے ہیں۔ انسان اور کمپیوٹر پانچ حسی راستوں سے بات کرتے ہیں: بصارت، چھونا، سننا، آواز اور مکانی حرکت۔ ہپٹکس (Haptics) وہ ارتعاش ہے جو آپ کو چھو کر محسوس ہوتا ہے۔

CHECK YOURSELF

1. Give two other names for HCI.
2. Which channel tracks body movement in VR?
3. Is a phone vibration sight, touch (haptics), or voice?

Answers: 1. CHI (Computer-Human Interface) and MMI (Man-Machine Interaction). 2. Spatial.
3. Touch / haptics.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. HCI studies:

- a. How people and computers work together
- b. Only CPU clocks
- c. Only NAND
- d. Only SQL keys

MCQ 2. Another name mentioned is:

- a. CHI / MMI
- b. MVP
- c. SDLC only
- d. K-map

MCQ 3. Sensory channels include:

- a. Sight, touch, hearing, voice, spatial
- b. Only smell in this course
- c. Only taste
- d. Only OSI layer 7

MCQ 4. Good HCI makes systems:

- a. Usable, efficient, fitting human needs
- b. Harder on purpose
- c. Mouse-only always
- d. Text in 4pt always

MCQ 5. Haptics are:

- a. Touch feedback (vibration)
- b. Pie charts
- c. Foreign keys
- d. Sprints

MCQ 6. VR tracking is a:

- a. Spatial channel
- b. Primary key
- c. Bubble sort
- d. Histogram

Extra short questions

1. Define HCI.
2. Five sensory channels from the notes.
3. Give a computer response for touch and for voice.
4. Why HCI is not "just drawing a pretty screen".

Extra long question

1. Tabulate the five channels with a human action and a computer response. Then explain one device that uses three channels at once.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Human-Computer Interaction: design of technology around people.
2. Sight, touch, hearing, voice, spatial.
3. Touch: click/haptic. Voice: speech recognition and spoken reply.
4. It includes users, tasks, environment, feedback, accessibility.

Long

1. Smartphone: sight (GUI), touch (tap/haptic), hearing (ringtone), optional voice assistant. Closing: if one channel fails, another should still work (accessibility).

Next lecture: 1.1.1 – 1.1.2 Traditional vs Natural Interaction